8 In this question you must show detailed reasoning.

(a) Find the centre and radius of the circle with equation $x^2 + y^2 - 2x + 4y - 20 = 0$. **[4]**

(b) Find the points of intersection of the circle with the line $x + 3y - 10 = 0$. **[5]**

8(a)

8(b)

(answer space continued on next page)

8(b)	**(continued)**